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Project Report

On

"Development of a Vision-Assisted Industrial Robot"

Submitted in partial fulfillment of the requirements for the
III semester

Mini Project Work[213ECE2442]

Bachelor of Engineering In

Electronics and Communication Engineering

Of

Kalasalingam Academy of Research and Education

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2025-2026

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CERTIFICATE

It is Certified that **Ms. ARAADHANA R, Ms. ABINAYA S, Mr. KATHIRVEL C and Ms. AFIYA RIFTHA O A** bearing REG NUM: 9925005014,9925005004, 9925005019, 9925005007 respectively, are Bonafide students of Kalasalingam Academy of Research and Education, and have completed requirements of the project entitled "Development of a Vision-Assisted Industrial Robot" partial fulfillment of the requirements for IV semester Bachelor of Engineering in Electronics and Communication Engineering during academic year 2025-26. It is certified that all Corrections/Suggestions indicated for Project Assessment have been in corporate in the report. The project report has been approved as it satisfies the academic requirements in respect of project work Phase-2 prescribed for the Bachelor of Engineering degree.

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1.

2.

DECLARATION

It is Certified that **Ms. ARAADHANA R, Ms. ABINAYA S, Mr. KATHIRVEL C and Ms. AFIYA RIFTHA O A** bearing REG NUM: 9925005014,9925005004, 9925005019, 9925005007 ,are bonafide students of Kalasalingam Academy of Research and Education, and have completed requirements of the project entitled "**Development of a Vision-Assisted Industrial Robot"** partial fulfillment of the requirements for IV semester Bachelor of Engineering in Electronics and Communication Engineering during academic year 2025-26.

We also declare that, to the best of our knowledge and belief, the work reported here does not form part of any other report on the basis of which a degree or award was conferred on an earlier occasion on this by any other student.

Date:

Place:

ABSTRACT

The project titled **"Development of a Vision-Assisted Industrial Robot"** focuses on developing an automated robotic system capable of detecting objects and performing pick-and-place operations with minimal human intervention. The proposed system combines computer vision technology and an STM32-based embedded control system to improve the efficiency, accuracy, and flexibility of industrial material-handling operations.

In traditional industrial environments, object identification and material handling often involve manual operations, which may require more time and effort. Automation using vision-assisted robotic systems can help overcome these limitations by enabling objects to be identified and handled systematically. The proposed system uses a camera to capture images of objects placed within the working area. The captured images are processed using suitable image-processing techniques to identify objects and estimate their positions.

The processed position information is communicated to the STM32 microcontroller, which acts as the main control unit of the robotic system. The microcontroller is programmed using Embedded C through STM32CubeIDE. Timer peripherals and Pulse Width Modulation (PWM) signals are used to control the servo motors responsible for the movement of the robotic arm. The arm consists of multiple joints, including the base, shoulder, and elbow, along with a gripper mechanism for holding and releasing objects.

Based on the received position information, the control system determines the required movement of the robotic arm. The arm moves towards the target object, operates the gripper to hold the object, transfers it to the designated location, and releases it. This sequence enables automated object handling and can be extended to support object sorting, packaging, and manufacturing operations.

The current development phase focuses on STM32 firmware development, timer configuration, PWM-based motor-control logic, and software simulation. STM32CubeIDE is used for program development and compilation, while Proteus is intended for circuit simulation, subject to compatible STM32 model availability. Physical hardware implementation and camera-based object detection are considered future development stages.

Overall, the proposed project demonstrates the application of embedded systems, computer vision, and robotic automation in industrial material handling. It provides a foundation for developing a flexible and efficient robotic system that can reduce manual effort, improve operational consistency, and support the advancement of smart industrial automation.

Contents

Chapter 1: Introduction

Chapter 2: Case study

Chapter 3: Methodology

Chapter4: Literature Survey

Chapter 5: Implementation

Chapter 6: Flow chart/Algorithm

Chapter 7: Simulation/result analysis

Chapter8: Conclusion and Future Scope

Chapter 9: References

CHAPTER-1

INTRODUCTION

The Vision-Assisted Industrial Robot is designed to automate object detection and pick-and-place operations in industrial applications. It combines computer vision technology with an STM32-based embedded control system to perform material-handling tasks with greater accuracy and efficiency.

A camera captures images of objects placed in the working area. The vision-processing system analyses the images to identify the objects and estimate their positions. This information is then provided to the STM32 microcontroller, which controls the robotic arm using servo motors.

The robotic arm moves according to the received position information, picks up the detected object using its gripper, and places it at the desired location. The STM32 is programmed using Embedded C to control the motor movements through PWM signals.

The main aim of this project is to reduce manual effort, minimize errors, and improve productivity in industrial environments. This system can be applied in manufacturing industries, assembly lines, packaging units, and automated sorting applications.

ProblemStatement

Traditional industrial pick-and-place operations often require manual effort for object identification, handling, and placement. These processes can be time-consuming, less accurate, and may lead to human errors. Therefore, there is a need for an automated system that can detect objects and perform accurate material-handling operations with minimal human intervention.

The proposed Vision-Assisted Industrial Robot aims to overcome these challenges by integrating a camera-based vision system with an STM32 microcontroller-controlled robotic arm. The system detects object positions and controls the robotic arm to pick and place objects at the required locations, improving accuracy, productivity, and efficiency in industrial applications.

Objectives

1. To develop an automated robotic system for industrial pick-and-place operations.
2. To detect and identify objects using a camera-based vision system.
3. To estimate the position of objects using image processing techniques.
4. To use an STM32 microcontroller for controlling the robotic arm.
5. To control servo motor movements using PWM signals generated through Embedded C programming.
6. To improve object-handling accuracy and reduce manual effort.
7. To demonstrate an efficient and cost-effective solution for industrial automation.

Scope of the Project

The project focuses on developing an automated pick-and-place robotic system using computer vision and an STM32 microcontroller. It detects object positions and controls the robotic arm to pick and place objects accurately. The system can be used in manufacturing, packaging, assembly lines, and industrial sorting applications.

Organization of the Report

This report is organized into different chapters to explain the development of the Vision-Assisted Industrial Robot.

- Chapter 1 – Introduction: Provides an overview, problem statement, objectives, and scope of the project.
- Chapter 2 – Literature Review: Discusses existing systems and related research works.
- Chapter 3 – System Design: Explains the block diagram, system architecture, and components used.
- Chapter 4 – Hardware and Software Implementation: Describes the STM32 microcontroller, programming, and robotic arm control.
- Chapter 5 – Simulation and Results: Presents the simulation setup, working process, and expected results.
- Chapter 6 – Conclusion and Future Scope: Summarizes the project and discusses possible improvements.

CHAPTER-2

CASESTUDY

Vision-Assisted Industrial Robotics in Manufacturing

Industrial manufacturing involves several repetitive operations, such as object identification, sorting, material handling, and product placement. Manual handling of these tasks can consume time and may result in positioning errors. Automated robotic systems help improve productivity, accuracy, and consistency in industrial processes.

Computer Vision and Industrial Automation

Computer vision enables robotic systems to identify objects and determine their positions using camera-based image processing. When integrated with embedded controllers, it allows robots to perform tasks based on the detected object information.

In this project, computer vision supports:

- Object Detection: Identifying objects in the working area using a camera.
- Position Estimation: Determining the location of detected objects.
- Automated Movement: Controlling robotic arm movements according to object positions.
- Pick-and-Place Operation: Using a gripper to pick objects and place them at the required locations.
- Embedded Control: Using an STM32 microcontroller to generate PWM signals for servo motor control.

Case: STM32-Based Vision-Assisted Pick-and-Place Robotic System

This project implements a vision-assisted robotic system using a camera, an STM32 microcontroller, and servo motors to demonstrate automated object handling.

Parameter	Always-OnBaseline	SmartIoTControl
Object Identification	Human observation	Camera based detection
Position Detection	Manual estimation	Image processing
Arm movement	Manual handling	STM32-controlled servos
Object gripping	Human-operated	Automated gripper
Object placement	Manual positioning	Programmed positioning

Monitoring	Human observation	System-based operation

Impact and Expected Results

The proposed system is expected to provide the following benefits:

- Reduced human effort in repetitive pick-and-place operations.
- Improved object-handling accuracy and consistency.
- Faster and more efficient material handling.
- Reduced manual positioning errors.
- Better integration of computer vision and embedded control.
- Potential use in manufacturing, packaging, assembly lines, and sorting industries.

CHAPTER–3

METHODOLOGY

SYSTEM ARCHITECTURE

The proposed system follows a camera-based vision and STM32-controlled robotic architecture. It consists of a vision-processing unit, an STM32 microcontroller, servo motor drivers, and a robotic arm mechanism. The camera captures images of objects, and the vision-processing system identifies their positions. The processed position information is then transferred to the STM32 controller, which controls the robotic arm to perform pick-and-place operations.

HARDWARE ARCHITECTURE

Vision Processing Unit

- Camera Module: Captures images of objects placed in the working area.
- Image Processing System: Processes the captured images to detect objects and estimate their positions.
- Communication Interface: Transfers the detected object position information to the STM32 controller.

STM32-Based Control Unit

- Microcontroller: STM32 development board.
- Programming Language: Embedded C using STM32 HAL libraries.
- PWM Outputs: Generates PWM signals to control the servo motors.
- GPIO Pins: Used for input signals and control operations.
- Power Supply: Provides the required regulated voltage to the controller and motor system.

Robotic Arm Unit

- Base Servo Motor: Controls the rotational movement of the robotic arm.
- Shoulder Servo Motor: Controls the vertical movement of the arm.
- Elbow Servo Motor: Adjusts the arm's reach and position.
- Gripper Servo Motor: Opens and closes the gripper to hold and release objects.
- Mechanical Structure: Supports the motors and robotic arm assembly.

SOFTWARE DESIGN

Vision Processing Software

- `image_capture()`: Captures images of objects using the camera.
- `object_detection()`: Identifies objects in the captured images.
- `position_estimation()`: Determines the approximate coordinates of detected objects.
- `data_transmission()`: Sends the processed position information to the STM32 controller.

STM32 Firmware Design

- `system_init()`: Initializes GPIO pins, timers, PWM channels, and other required peripherals.
- `receive_position()`: Receives object position information from the vision-processing unit.
- `servo_control()`: Generates PWM signals to control the servo motors.
- `robotic_sequence()`: Executes the programmed robotic arm movement sequence.
- `gripper_control()`: Controls the gripping and releasing operations.
- `system_status()`: Monitors the operating sequence and control status.

ROBOTIC ARM CONTROL ALGORITHM

The STM32 microcontroller controls the robotic arm based on the received object position information.

The operation follows these steps:

1. The camera captures the image of the object.
2. The image-processing system detects the object and estimates its position.
3. The position information is transmitted to the STM32 microcontroller.
4. The STM32 calculates the required servo motor movements.
5. The robotic arm moves towards the detected object.
6. The gripper closes and picks up the object.
7. The arm moves towards the target location.
8. The gripper releases the object at the designated position.
9. The robotic arm returns to its initial position.

PWM-BASED MOTOR CONTROL

The servo motors are controlled using Pulse Width Modulation (PWM) signals generated by the STM32 microcontroller.

- PWM Frequency: Typically 50 Hz for standard hobby servos.
- Pulse Width: Approximately 1–2 ms for a typical servo operating range.
- Motor Position: Controlled by varying the PWM pulse width.
- Motor Coordination: Multiple PWM channels control the base, shoulder, elbow, and gripper movements.

This allows the robotic arm to perform controlled and coordinated movements during pick-and-place operations.

DASHBOARD / MONITORING ARCHITECTURE

If a monitoring interface is included in the project, it can display the operating status of the robotic system.

- Object Detection Status: Displays whether an object has been detected.
- Position Information: Displays the estimated object coordinates.
- Robot Status: Indicates whether the robotic arm is idle or performing an operation.
- Motor Status: Displays the movement status of the servo motors.
- Operation Counter: Maintains the number of completed pick-and-place cycles.

EXPECTED SYSTEM OUTPUT

The proposed system is expected to detect objects using computer vision and automatically control the robotic arm through the STM32 microcontroller. It performs object picking, transferring, and placement with reduced human intervention.

CHAPTER – 4

LITERATURESURVEY

Computer Vision in Industrial Automation

Szeliski (2010), in Computer Vision: Algorithms and Applications, explained various image-processing techniques used for object detection, recognition, and position estimation. These techniques provide the foundation for developing vision-assisted robotic systems that can identify objects and determine their locations.

Bajcsy et al. (2018) discussed the importance of visual perception in robotic applications, highlighting how camera-based information helps robots interact with objects and perform automated tasks.

Vision-Assisted Robotic Systems

Research on vision-guided robotics demonstrates that integrating cameras with robotic arms improves object identification and positioning accuracy. Camera-based systems allow robots to identify objects in their working environment and adjust their movements according to the detected positions.

This approach forms the basis of the proposed vision-assisted pick-and-place system.

STM32 Microcontroller in Embedded Robotics

The STM32 family of microcontrollers is widely used in embedded applications due to its processing capabilities, GPIO interfaces, timers, and PWM peripherals. These features make STM32 suitable for controlling servo motors and coordinating robotic arm movements.

STM32-based embedded systems provide a flexible platform for implementing real-time motor-control operations using Embedded C programming.

PWM-Based Servo Motor Control

Pulse Width Modulation (PWM) is commonly used to control the angular position of servo motors in robotic applications. By varying the pulse width, the microcontroller can control the movement of individual robotic arm joints.

This technique is used in the proposed system to control the base, shoulder, elbow, and gripper movements.

Robotic Arm and Pick-and-Place Automation

Industrial robotic arms are commonly used for material handling, sorting, assembly, and packaging operations. Pick-and-place robots reduce repetitive manual work and improve consistency in object handling.

The proposed project uses a servo-controlled robotic arm and gripper to perform object picking, transferring, and placement based on detected object positions.

Embedded Programming for Robotic Applications

Embedded C programming provides direct control over microcontroller peripherals such as timers, GPIO pins, and PWM channels. STM32 HAL libraries simplify peripheral configuration and motor-control implementation.

The project uses STM32CubeIDE and Embedded C to develop the firmware required for controlling the robotic arm.

Applications of Vision-Assisted Robotics

Vision-assisted robotic systems are increasingly useful in manufacturing industries because they combine object recognition with automated movement. Such systems can be applied in:

- Automated material handling.
- Product sorting and classification.
- Industrial assembly lines.
- Packaging and manufacturing units.
- Object inspection and placement.

Research Gap Addressed

Many traditional pick-and-place systems depend on fixed positions or manual object placement. Such systems have limitations when objects are placed at different positions.

The proposed project addresses this limitation by integrating a camera-based vision system with an STM32-controlled robotic arm. It aims to detect object positions and perform automated pick-and-place operations, providing a foundation for low-cost vision-assisted industrial automation.

CHAPTER-5

IMPLEMENTATION

System Overview

The implementation of the Vision-Assisted Industrial Robot is currently focused on software development and simulation. The system consists of STM32 firmware developed using STM32CubeIDE and a robotic control circuit designed using Proteus simulation software.

The simulation demonstrates the working of the robotic arm, servo motor control, and automated pick-and-place sequence. Physical hardware implementation and camera integration are planned for the later phase of the project.

STM32 Firmware – Key Implementation Details

Microcontroller Configuration

The STM32 microcontroller is configured to control the robotic arm through GPIO and PWM peripherals.

- Microcontroller: STM32 development board.
- Programming Language: Embedded C.
- Development Environment: STM32CubeIDE.
- Simulation Software: Proteus.
- Control Method: PWM-based servo motor control.

GPIO Configuration

GPIO pins are configured to receive input signals and control the robotic operation.

- Input signals are used to initiate the pick-and-place sequence.
- GPIO outputs are configured for required control operations.
- Timer peripherals are configured to generate PWM signals.

PWM Configuration

PWM signals are generated using STM32 timer peripherals to control the servo motor positions.

- PWM Frequency: Typically 50 Hz for standard servo motors.
- Pulse Width: Approximately 1–2 ms.
- Motor Control: Different pulse widths produce different servo angles.
- Application: Controls the base, shoulder, elbow, and gripper movements.

Robotic Arm Control Implementation

The robotic arm movement is programmed using Embedded C. The firmware executes a predefined sequence to demonstrate object picking and placement.

Action	Base	Shoulder	Elbow	Gripper
Initial position	90	60	45	20
Move to pick	90	80	70	20
Grab object	90	80	70	60
Lift object	90	40	30	60
Rotate to drop	140	40	30	60
Release object	140	40	30	20

Proteus Simulation Implementation

Proteus is used to design and simulate the electronic control circuit.

Simulation Steps

1. Create a new project in Proteus.
2. Add the supported STM32 microcontroller model.
3. Place the servo motor components in the schematic.
4. Connect the servo signal pins to the configured STM32 PWM outputs.
5. Connect the required power and ground terminals.
6. Generate the firmware HEX file using STM32CubeIDE.
7. Load the HEX file into the STM32 simulation component.
8. Configure the microcontroller clock frequency.
9. Run the simulation and observe the servo motor control signals.
10. Verify the programmed pick-and-place sequence.

Software Implementation

STM32CubeIDE

STM32CubeIDE is used for developing and compiling the Embedded C program.

- Configures GPIO pins and timer peripherals.
- Generates PWM signals for servo motor control.
- Implements the robotic arm movement sequence.
- Builds and generates the firmware output file.
- Helps debug and verify the embedded program.

Proteus Design Suite

Proteus is used to represent the circuit and verify the microcontroller-based control operation.

- Simulates the STM32 control circuit, provided a compatible VSM model is available.
- Displays PWM signal outputs.
- Helps verify servo motor control.
- Supports testing of the programmed robotic sequence.

Expected Simulation Output

The simulation is expected to demonstrate the following operations:

- Initialization of the STM32 microcontroller.
- Generation of PWM signals for servo motors.
- Individual movement of robotic arm joints.
- Opening and closing of the gripper.
- Execution of the complete pick-and-place sequence.
- Return of the robotic arm to its initial position.

Future Hardware Implementation

After completing the software simulation, the project will proceed to physical implementation. This phase will include STM32 board integration, camera interfacing, servo motor connections, robotic arm assembly, and real-time object detection.

CHAPTER-6

FLOWCHART/ALGORITM

Overall System Algorithm

The proposed Vision-Assisted Industrial Robot follows a sequence of operations involving object detection, position estimation, STM32-based control, and robotic arm movement. The current implementation focuses on software development and simulation.

1. System Initialization

1. Start the STM32 microcontroller.
2. Initialize the required GPIO pins.
3. Configure timer peripherals for PWM generation.
4. Initialize servo motor control channels.
5. Set the robotic arm to its initial position.
6. Initialize the input signals for starting the operation.
7. Wait for the object detection or start command.

2. Object Detection and Position Estimation

8. Capture the object image using the camera (planned for the hardware phase).
9. Process the captured image using the vision-processing system.
10. Detect and identify the object.
11. Estimate the position of the detected object.
12. Transmit the position information to the STM32 controller.

3. STM32 Robotic Control Algorithm

13. Receive the object position information.
14. Check whether valid position information is available.
15. Calculate the required robotic arm movement.
16. Generate PWM signals for the servo motors.
17. Move the robotic arm towards the object.
18. Activate the gripper to hold the object.
19. Lift the object from its initial position.
20. Move the arm towards the target location.
21. Release the object at the required position.
22. Return the robotic arm to its initial position.
23. Repeat the process for the next object.

Servo Motor Control Algorithm

1. Configure the STM32 timer for PWM generation.
2. Set the required PWM frequency.
3. Assign PWM channels to the servo motors.
4. Define the required angular positions.
5. Convert the desired angles into corresponding PWM pulse widths.
6. Generate PWM signals for each servo motor.
7. Update the motor positions according to the pick-and-place sequence.
8. Stop or hold the motors at their final positions.

Simulation Algorithm

1. Design the control circuit in Proteus.
2. Develop the STM32 Embedded C program in STM32CubeIDE.
3. Configure the GPIO and PWM peripherals.
4. Compile the program and generate the HEX file.
5. Load the HEX file into the supported STM32 simulation model.
6. Run the simulation.
7. Observe the PWM outputs and servo motor movements.
8. Verify the programmed pick-and-place sequence.
9. Record the simulation results.

SOFTWARE AND HARDWARE USED:

CODE:1

```
  void Servo_SetAngle(uint32_t channel, uint8_t angle)
{
   uint32_t pulse = 1000 + ((uint32_t)angle * 1000) / 180;

   __HAL_TIM_SET_COMPARE(&htim2, channel, pulse);
}


HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_1);
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_2);
HAL_TIM_PWM_Start(&htim2, TIM_CHANNEL_3);

Servo_SetAngle(TIM_CHANNEL_1, 90);
Servo_SetAngle(TIM_CHANNEL_2, 90);
Servo_SetAngle(TIM_CHANNEL_3, 20);



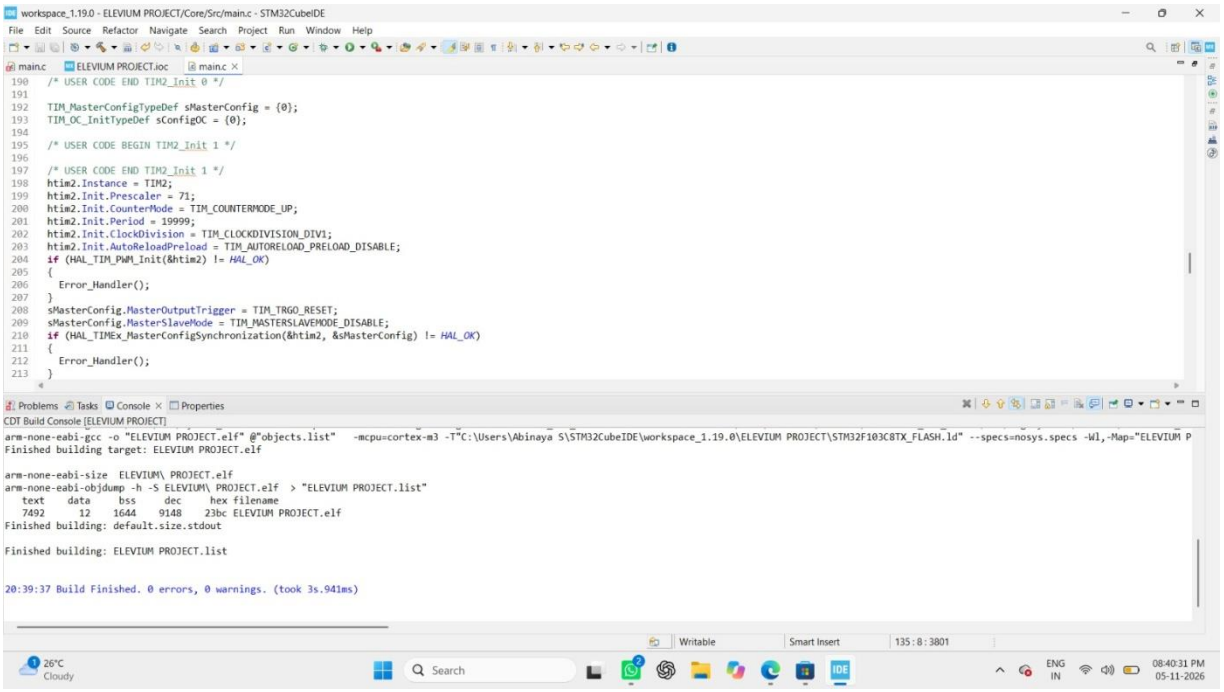
if (HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_8) == GPIO_PIN_SET &&
  HAL_GPIO_ReadPin(GPIOA, GPIO_PIN_9) == GPIO_PIN_SET)
{
  // Move arm towards the object
  Servo_SetAngle(TIM_CHANNEL_1, 45);
  Servo_SetAngle(TIM_CHANNEL_2, 120);
  HAL_Delay(1000);

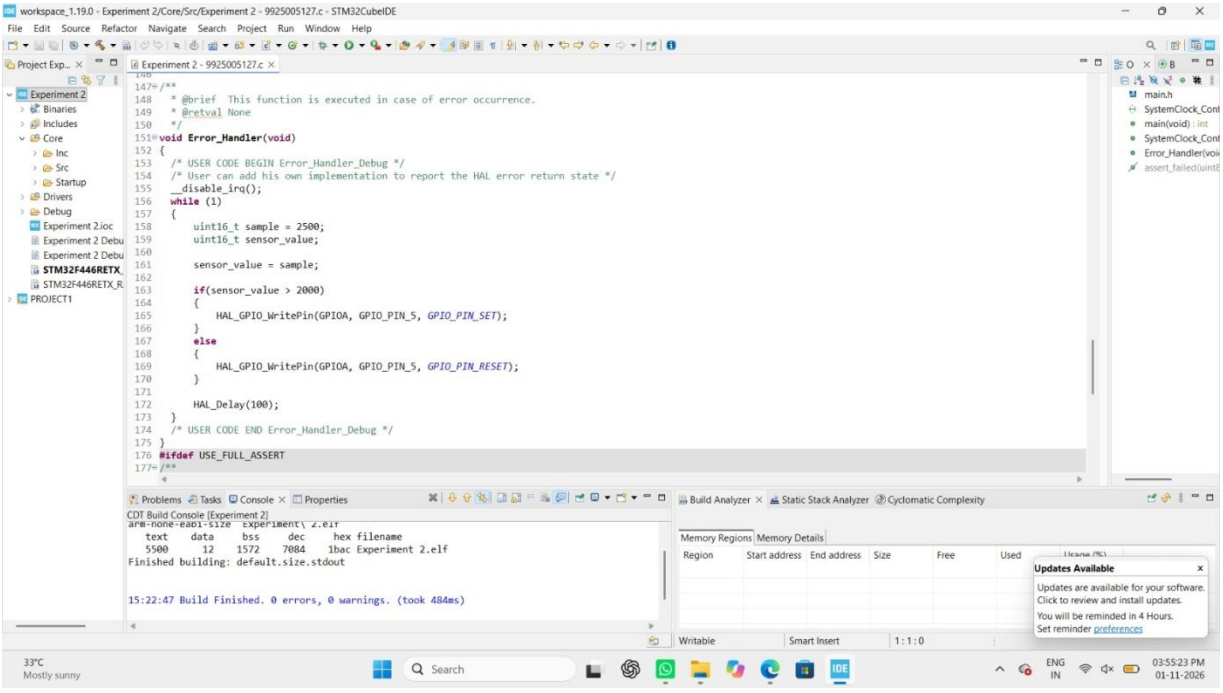
  // Close gripper
  Servo_SetAngle(TIM_CHANNEL_3, 90);
  HAL_Delay(1000);

  // Move arm to release position
  Servo_SetAngle(TIM_CHANNEL_1, 135);
  Servo_SetAngle(TIM_CHANNEL_2, 60);
  HAL_Delay(1000);

  // Open gripper
  Servo_SetAngle(TIM_CHANNEL_3, 20);
  HAL_Delay(1000);

  // Return to initial position
  Servo_SetAngle(TIM_CHANNEL_1, 90);
  Servo_SetAngle(TIM_CHANNEL_2, 90);
}
```

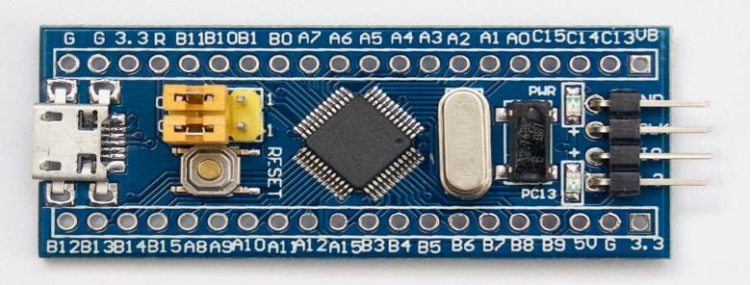

workspace_1.19.0 - ELEVIUM PROJECT/Core/Src/main.c - STM32CubeIDE
File Edit Source Refactor Navigate Search Project Run Window Help
main.c
ELEVIUM PROJECT.ioc
main.c
/* USER CODE END TIM2_Init 0 */
TIM_MasterConfigTypeDef sMasterConfig = {0};
TIM_OC_InitTypeDef sConfigOC = {0};
/* USER CODE BEGIN TIM2_Init 1 */
/* USER CODE END TIM2_Init 1 */
htim2.Instance = TIM2;
htim2.Init.Prescaler = 71;
htim2.Init.CounterMode = TIM_COUNTERMODE_UP;
htim2.Init.Period = 19999;
htim2.Init.ClockDivision = TIM_CLOCKDIVISION_DIV1;
htim2.Init.AutoReloadPreload = TIM_AUTORELOAD_PRELOAD_DISABLE;
if (HAL_TIM_PWM_Init(&htim2) != HAL_OK)
{
Error_Handler();
}
sMasterConfig.MasterOutputTrigger = TIM_TRGO_RESET;
sMasterConfig.MasterSlaveMode = TIM_MASTERSLAVEMODE_DISABLE;
if (HAL_TIMEx_MasterConfigSynchronization(&htim2, &sMasterConfig) != HAL_OK)
{
Error_Handler();
}
Problems Tasks Console Properties
CDT Build Console [ELEVIUM PROJECT]
arm-none-eabi-gcc -o "ELEVIUM PROJECT.elf" @"objects.list" -mcpu=cortex-m3 -T"C:\Users\Abinaya S\STM32CubeIDE\workspace_1.19.0\ELEVIUM PROJECT\STM32F103C8TX_FLASH.ld" --specs=nosys.specs -Wl,-Map="ELEVIUM P
Finished building target: ELEVIUM PROJECT.elf
arm-none-eabi-size ELEVIUM\ PROJECT.elf
arm-none-eabi-objdump -h -S ELEVIUM\ PROJECT.elf > "ELEVIUM PROJECT.list"
text data bss dec hex filename
7492 12 1644 9148 23bc ELEVIUM PROJECT.elf
Finished building: default.size.stdout
Finished building: ELEVIUM PROJECT.list
20:39:37 Build Finished. 0 errors, 0 warnings. (took 3s.941ms)
Writable Smart Insert 135 : 8 : 3801
26°C Cloudy
Search
ENG IN
08:40:31 PM 05-11-2026


workspace_1.19.0 - Experiment 2/Core/Src/Experiment 2 - 9925005127.c - STM32CubeIDE
File Edit Source Refactor Navigate Search Project Run Window Help
Project Exp...
Experiment 2
Binaries
Includes
Core
Inc
Src
Startup
Drivers
Debug
Experiment 2.ioc
Experiment 2 Debu
Experiment 2 Debu
STM32F446RETX_
STM32F446RETX_R
PROJECT1
Experiment 2 - 9925005127.c
/**
* @brief This function is executed in case of error occurrence.
* @retval None
*/
void Error_Handler(void)
{
/* USER CODE BEGIN Error_Handler_Debug */
/* User can add his own implementation to report the HAL error return state */
__disable_irq();
while (1)
{
uint16_t sample = 2500;
uint16_t sensor_value;
sensor_value = sample;
if(sensor_value > 2000)
{
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_SET);
}
else
{
HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, GPIO_PIN_RESET);
}
HAL_Delay(100);
}
/* USER CODE END Error_Handler_Debug */
}
#ifdef USE_FULL_ASSERT
/**
main.h
SystemClock_Conf
main(void) : int
SystemClock_Conf
Error_Handler(voi
assert_failed(uint8
Problems Tasks Console Properties
CDT Build Console [Experiment 2]
arm-none-eabi-size "Experiment 2.elf"
text data bss dec hex filename
5500 12 1572 7084 1bac Experiment 2.elf
Finished building: default.size.stdout
15:22:47 Build Finished. 0 errors, 0 warnings. (took 484ms)
Build Analyzer
Static Stack Analyzer
Cyclomatic Complexity
Memory Regions
Memory Details
Region
Start address
End address
Size
Free
Used
Updates Available
Updates are available for your software.
Click to review and install updates.
You will be reminded in 4 Hours.
Set reminder preferences
Writable
Smart Insert
1 : 1 : 0
33°C Mostly sunny
Search
ENG IN
03:55:23 PM 01-11-2026

HARDWARE

STM32 Microcontroller

The STM32 microcontroller acts as the main control unit of the robotic system. It receives object-position information and controls the servo motors through PWM signals. The controller is programmed using Embedded C.



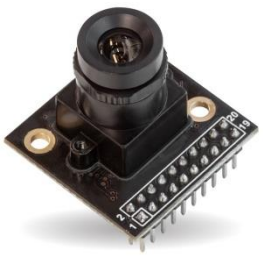
Robotic Arm

The robotic arm consists of interconnected links and joints that allow movement in different directions. It helps the system reach an object, lift it, and transfer it to the required location.



Camera Module

The camera captures images of objects in the working area. The images are analysed by the vision-processing system to identify objects and estimate their positions. Camera integration is planned for the hardware phase.



Servo Motors

Servo motors provide controlled angular movement to the robotic arm. They are used to control the base, shoulder, elbow, and gripper. Their positions are controlled using PWM signals from the STM32.



Power Supply

The power supply provides the required electrical energy to the microcontroller and servo motors. During hardware implementation, suitable regulated supplies will be selected according to their voltage andcurrentrequirements.



Connecting Wires

Connecting wires are used to establish electrical connections between the STM32 microcontroller, servo motors, and other electronic components.



CHAPTER – 7
STIMULATIONANDRESULT

Simulation Setup

The proposed Vision-Assisted Industrial Robot is currently developed and tested at the software level. STM32CubeIDE is used to write and compile the Embedded C program for the STM32F103C8Tx microcontroller. The program includes timer configuration and control logic required for the robotic arm operation.

The timer peripheral is configured using TIM2 with a prescaler value of 71 and a counter period of 19999. This configuration is intended to generate a 50 Hz PWM signal when the timer input clock is 72 MHz. The generated PWM signals will be used to control the servo motors of the robotic arm.

Proteus is planned to be used to simulate the microcontroller and servo motor control circuit, subject to the availability of a compatible STM32 simulation model.

Test	Expected Behaviour	Observed Result
STM32 project compilation	Program should compile successfully	PASS – Build completed
Compilation errors	No compilation errors	0 errors
Compilation warnings	No compilation warnings	0 warnings
TIM2 initialization	Timer should be configured successfully	Configuration present in code
Timer prescaler	Prescaler value should be 71	PASS – Value configured
Timer period	Counter period should be 19999	PASS – Value configured

PWM Configuration Validation

The TIM2 peripheral is configured to generate PWM signals for servo motor control. PWM is used to control the angular position of the robotic arm joints.

Parameter	Configured Value
Timer	TIM2
Prescaler	71
Counter period	19999
Counter mode	Up-counting
Intended PWM frequency	50 Hz*
Application	Servo motor control

Control Logic Validation

The program also contains conditional logic that compares a sensor value with a predefined threshold and controls GPIOA pin 5.

Test Condition	Expected Behaviour	Observed Result
Sensor value = 2500	Condition should be TRUE	TRUE
Sensor value greater than 2000	GPIOA pin 5 should be HIGH	HIGH
Sensor value less than or equal to 2000	GPIOA pin 5 should be LOW	LOW

Robotic Arm Simulation Validation

The proposed robotic arm simulation is intended to verify the movement of individual joints and the complete pick-and-place sequence.

Test Condition	Expected Behaviour	Result
System initialization	Robotic arm moves to its initial position	To be verified
Base servo activation	Base rotates to the required position	To be verified
Shoulder servo activation	Shoulder joint moves as programmed	To be verified
Elbow servo activation	Elbow joint moves as programmed	To be verified
Gripper activation	Gripper opens and closes	To be verified
Pick-and-place sequence	Arm follows the programmed movement sequence	To be verified
Sequence completion	Arm returns to its initial position	To be verified

Analysis

The STM32 firmware compilation was completed successfully using STM32CubeIDE, with zero errors and zero warnings. The TIM2 timer configuration provides the required foundation for generating PWM signals for servo motor control.

The conditional control logic demonstrates how sensor values can be compared against predefined thresholds to control GPIO outputs. These programming concepts can be extended to the robotic arm control system.

The next stage of simulation involves verifying individual servo movements, gripper operation, and the complete pick-and-place sequence. Camera-based object detection and physical robotic arm integration are planned for the future hardware phase.

CHAPTER – 8
CONCLUSION AND FUTURE SCOPE

Conclusion

The project "Development of a Vision-Assisted Industrial Robot" aims to develop an automated robotic system for object detection and pick-and-place operations in industrial applications. The proposed system combines computer vision technology with an STM32 microcontroller-based control system to improve the accuracy and efficiency of object handling.

The STM32 microcontroller is programmed using Embedded C in STM32CubeIDE. Timer configuration and PWM generation are used as the foundation for controlling the servo motors of the robotic arm. The initial firmware was compiled successfully with zero errors and zero warnings. The current simulation phase focuses on verifying the control logic, servo movements, and programmed pick-and-place sequence.

The project demonstrates the application of embedded systems and robotic automation in industrial material handling. It provides a foundation for developing a vision-assisted robotic system that can reduce manual effort, improve consistency, and support automated production processes.

Key achievements and development activities:

- Developed the initial STM32 firmware using Embedded C.
- Configured the TIM2 timer for the intended PWM-based servo control.
- Successfully compiled the STM32 program with zero errors and zero warnings.
- Developed basic conditional control logic using sensor-value thresholds.
- Planned the robotic arm control sequence for pick-and-place operations.
- Established a software and simulation approach for future system integration.

Future Scope

1. Camera Integration and Object Detection

A camera module can be integrated into the system to capture images of objects in the working area. Image-processing techniques can be used to identify objects and determine their positions. The detected position information can then be transmitted to the STM32 controller.

2. Advanced Image Processing

Advanced image-processing algorithms can be implemented to identify objects based on their shape, colour, size, and orientation. This can improve object recognition and make the robotic system suitable for different sorting applications.

3. AI-Based Object Recognition

Machine learning and deep learning techniques can be introduced to recognize different types of objects. This can help the robotic arm automatically classify and handle objects without requiring predefined positions for every object.

4. Real-Time Robotic Arm Implementation

The simulated control system can be implemented using a physical STM32 development board, servo motors, and a robotic arm structure. This will help verify the actual movement, gripping capability, and pick-and-place performance.

5. Improved Servo Motor Control

Advanced control techniques can be implemented to achieve smoother and more accurate movement of the robotic arm. Calibration and feedback mechanisms can also be introduced to improve positioning accuracy.

6. Wireless Communication

Wireless communication can be added to transfer object-position information between the image-processing system and the STM32 controller. This can provide flexibility in industrial applications.

7. Industrial Automation Integration

The system can be further developed for integration with industrial conveyor belts, automated sorting systems, and manufacturing assembly lines. This can support efficient material handling and reduce manual intervention.

8. Multi-Object Handling

The robotic arm can be upgraded to detect, classify, and transfer multiple objects to different locations. This can improve the flexibility and productivity of the system in industrial environments.

CHAPTER – 9
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